

Sydney Girls High School



MATHEMATICS EXTENSION 2

Year 12

Assessment Task 1

2023

Reading time : 5 minutes

Writing time : 60 minutes

Total marks : 50

Topics: Complex Numbers (partial topic), Proof (partial topics)

General Instructions:

- There are four (4) questions which **are not** of equal value.
- Attempt all questions.
- Start each question on a NEW page.
- Show all necessary working. Marks may be deducted for badly arranged work or incomplete working.
- A correct answer without working may be awarded a maximum of 1 mark.
- Write on one side of the paper only.
- Diagrams are NOT to scale, unless indicated otherwise.
- Board-approved calculators may be used.
- Write using a black or blue pen.
- White-out or correction tape is NOT to be used.
- A Mathematics reference sheet is provided.

Student Name: _____

Teacher Name: _____

Question 1 (13 marks)

- a) Write the negation of: $\exists$ a real number $x \in \mathbb{R}$ such that $x^2 - 25 = 0$. 1
- b) Given $z = 1 - 2i$ and $w = 1 + i$. Evaluate:
- i) $z + w$ 1
- ii) $\frac{z}{w}$ 2
- c) Prove by counterexample that the statement $x^2 + y^2 < 2xy$ for all $x, y \in \mathbb{R}$ is false. 2
- d) Convert $\sqrt{2} \left(\cos \left(\frac{-\pi}{4} \right) + i \sin \left(\frac{\pi}{4} \right) \right)$ into cartesian form. 2
- e) Prove by contrapositive that: if $3n + 2$ is even, then n is even, given $n \in \mathbb{Z}^+$. 2
- f) Factorise $z^2 + 6iz + 7$. Hence solve $z^2 + 6iz + 7 = 0$. 3

End of Question 1

Question 2 (12 marks) Begin writing on a NEW page.

- a) Convert $\frac{1}{1+i}$ into modulus argument form. 2
- b) Prove that the sum of any four consecutive numbers is always even. 2
- c) Given: $z = 2 - 3i$ is a zero of $p(z) = z^3 + z^2 - 7z + 65$, find the other two zeros. 2
- d) Find the square roots of the complex number $7 + 24i$. 3
- e) Given: $a^m - b^m = (a - b)(a^{m-1} + a^{m-2}b + a^{m-3}b^2 + \dots + b^{m-1})$. Prove $3^{2n} - 1$ is divisible by 4 without using mathematical induction. 3

End of Question 2

Question 3 (13 marks) Begin writing on a NEW page.

- a) On the Argand diagram, let A represent $z_1 = 1 + i\sqrt{3}$, let B represent $z_2 = 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$ and let C represent $z_3 = z_1 + z_2$. Find the exact coordinates of point C . 1
- b) Prove the following statement is false:
“There is a Pythagorean Triad where all numbers are odd.” 2
- c) Sketch the following regions, showing all important features. You do not need to include axis intercepts.
- i) $|z + i| \leq |z - 2|$ 2
- ii) $-\frac{\pi}{4} < \text{Arg}(z + 1 - i\sqrt{5}) \leq \frac{\pi}{4}$. 2
- d) Let A and B be points in the complex plane representing $2i$ and $6 + 4i$ respectively. Suppose AB is one side of a square $ABCD$. Find all the possible complex numbers representing C and D . 3
- e) Use proof by contradiction to show that $\frac{1}{\sqrt{2}}$ is irrational. 3

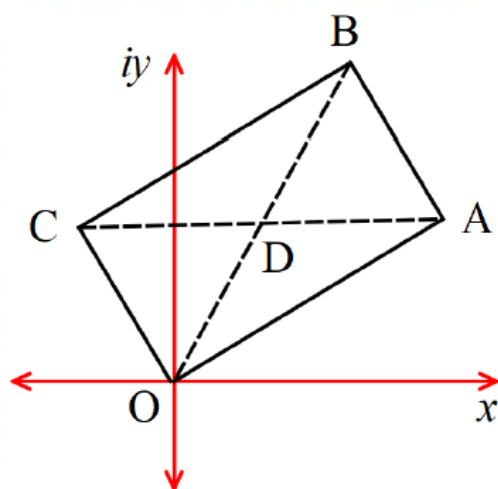
End of Question 3

Question 4 (12 marks) Begin writing on a NEW page.

- a) Prove that a three-digit number is divisible by 9 if and only if the sum of its digits is divisible by 9. 3

- b) In the Argand diagram, $OABC$ is a rectangle, where $OA = 2OC$. 2
 The vertex A corresponds to the complex number w .
 The diagonals OB and AC intersect at D .

Find the complex number that represents D in terms of w .



- c) Prove that $4^n + 111(2^n) + 2024$ is composite for all $n \in \mathbb{Z}^+$. 2

- d) z_1 , z_2 and z_3 are three complex numbers which satisfy $z_1 + z_2 + z_3 = 0$. 2

- i) Given that the arguments of z_1 , z_2 and z_3 are α , β and γ respectively,
 and their moduli are 1, k and $2-k$ respectively, where $0 < k < 2$.
 Express z_1 , z_2 and z_3 in modulus-argument form.

- ii) Prove that: $k^2 + (2-k)^2 + 2k(2-k)\cos(\beta - \gamma) = 1$. 3

End of Task.

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Q1

e) prove by contrapositive:

If n is Not even (odd) then $3n+2$ is Not even (odd).

Let $n = 2k-1$, $k \in \mathbb{Z}^+$

$$3n + 2 = 3(2k-1) + 2$$
$$= 6k - 1$$

$$= 2(3k) - 1$$

$$= 2p - 1, p \in \mathbb{Z}^+$$

This is odd.

* One mark will be deducted for letting $n = 2k+1$, $k \in \mathbb{Z}^+$, as this will leave out 1 as an odd number.

f) $z^2 + 6iz + 7$

$$(z + 7i)(z - i) \checkmark$$

$$z^2 + 6iz + 7 = 0$$

$$(z + 7i)(z - i) = 0$$

$$\begin{cases} z = -7i \checkmark \\ z = i \checkmark \end{cases}$$

Students must factorise

$z^2 + 6iz + 7$ into $(z + 7i)(z - i)$ then solve for z to get 3 Marks.

Q1

a) $\forall x \in \mathbb{R}, x^2 - 25 \neq 0$

b) $z = 1 - 2i, w = 1 + i$

i) $z + w = 1 - 2i + 1 + i$
 $= 2 - i$

ii) $\frac{z}{w} = \frac{1 - 2i}{1 - i} \times \frac{(1 + i)}{(1 + i)}$
 $= \frac{1 + i - 2i + 2}{1 + 1} = \frac{3}{2} - \frac{1}{2}i$

c) $x^2 + y^2 < 2xy, x, y \in \mathbb{R}$

Counterexample: For $x = 1, y = 1$

LHS = $1^2 + 1^2 = 2$

RHS = $2(1)(1) = 2$

LHS = RHS = 2 $\therefore$ LHS $\nless$ RHS

$\therefore x^2 + y^2 < 2xy$ is false.

OR any logical values of x, y that disproves the original statement.

d) $\sqrt{2} \left(\cos\left(-\frac{\pi}{4}\right) + i \sin\left(\frac{\pi}{4}\right) \right)$

$= \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$

$= \sqrt{2} \left(\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}} \right)$

$= 1 + i$

Marks Question 2

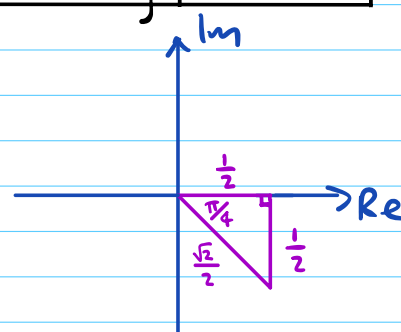
2 (a)

Criteria	Marks
• Provides correct solution	2
• Finds $\frac{1}{1+i} = \frac{1}{2} - \frac{1}{2}i$ OR	1
• Finds modulus or argument correctly	

$$\frac{1}{1+i} \times \frac{1-i}{1-i}$$
$$= \frac{1-i}{1^2 + 1^2}$$

$$= \frac{1}{2} - \frac{1}{2}i \quad \checkmark$$

$$= \frac{1}{\sqrt{2}} \operatorname{cis} \left(-\frac{\pi}{4} \right) \quad \checkmark$$



$$\left| \frac{1}{2} - \frac{1}{2}i \right| = \sqrt{\left(\frac{1}{2} \right)^2 + \left(\frac{1}{2} \right)^2}$$
$$= \sqrt{\frac{1}{4} + \frac{1}{4}}$$
$$= \sqrt{\frac{1}{2}}$$
$$= \frac{1}{\sqrt{2}}$$

Comments:

- Students who sketched the location of $\frac{1}{2} - \frac{1}{2}i$ were more successful than those using inverse tan with no sketch.

Marks

2

Question 2 (continued)

(b)

Criteria	Marks
• Provides correct solution	2
• Calculates $n + (n+1) + (n+2) + (n+3) = 4n + 6$ or equivalent merit	1

Let the four consecutive numbers be $n, n+1, n+2, n+3$.

∴ Their sum is

$$n + n+1 + n+2 + n+3 = 4n + 6$$

$$= 2(2n + 3)$$

which is even, as required.

Comments:

- Some students split this question into cases depending on whether the first number was odd or even. While this is often a useful strategy, it was not necessary here.
- Some students wrote $2n, 2n+1, 2n+2, 2n+3$, which incorrectly assumes the first number is even.

Marks

2

Question 2 (continued)

(c)

Criteria	Marks
• Provides correct solution	2
• Applies the conjugate root theorem OR	1
• Considers the sum or product of roots	

Since all of the coefficients are real,
the conjugate root theorem applies.

Hence, $\overline{2-3i} = 2+3i$ is also a root. ✓

Let the remaining root be α .

Therefore, the sum of roots is

$$2 + \cancel{3i} + 2 - \cancel{3i} + \alpha = -\frac{b}{a}$$

$$4 + \alpha = -1$$

$$\alpha = -5 \quad \checkmark$$

$\therefore$ The other two roots are $2+3i$ and -5 .

Comments:

- This question was very well done.

The majority of students applied
the conjugate root theorem and sum
of roots to efficiently find all roots.

Marks

3

Question 2 (continued)

(d)

Criteria	Marks
• Provides correct solution	3
• Attempts to solve $x^2 - y^2 = 7$ and $2xy = 24$ OR	2
• Finds both square roots without working	
• Sets up $(x + iy)^2 = 7 + 24i$ OR	1
• Finds one square root without working	

We want $x + iy$ such that

$$(x + iy)^2 = 7 + 24i$$

$$x^2 - y^2 + 2xyi = 7 + 24i \quad \checkmark$$

Equate real and imaginary parts:

$$x^2 - y^2 = 7$$

$$2xy = 24 \quad \checkmark$$

$$xy = 12$$

By inspection, $x = \pm 4$, $y = \pm 3$

$\therefore$ The square roots of $7 + 24i$

are $4 + 3i$ and $-4 - 3i \quad \checkmark$

Comments:

- Quite well done. Some working out was required for full marks.

Marks Question 2 (continued)

(e)

3

Criteria	Marks
• Provides correct solution	3
• Progress proving $3^{2n-1} + 3^{2n-2} + \dots + 3 + 1$ is even OR	2
• Writes $q^n - 1 = (q-1)(q^{n-1} + q^{n-2} + \dots + 1)$ OR	
• Progress proving $(3^n - 1)(3^n + 1)$ divisible by 4 OR	
• Writes $3^{2n} - (-1)^{2n} = (3+1)(3^{2n-1} + 3^{2n-2}(-1) + \dots + (-1)^{2n-1})$	
• Writes $3^{2n} - 1 = (3-1)(3^{2n-1} + 3^{2n-2} + \dots + 3 + 1)$ OR	1
• Writes $3^{2n} - 1 = q^n - 1$ OR	
• Writes $3^{2n} - 1 = (3^n - 1)(3^n + 1)$ OR	
• Writes $3^{2n} - 1 = 3^{2n} - (-1)^{2n}$	

$$\begin{aligned}
 3^{2n} - 1 &= q^n - 1 \quad \checkmark \\
 &= (q-1)(q^{n-1} + q^{n-2} + \dots + q + 1) \quad \checkmark \\
 &= 8(q^{n-1} + q^{n-2} + \dots + q + 1) \\
 &= 4(2 \times q^{n-1} + 2 \times q^{n-2} + \dots + 2)
 \end{aligned}$$

which is divisible by 4. $\checkmark$

Note: This method proves the stronger statement that $3^{2n} - 1$ is divisible by 8.

Comments:

- As you can see from the marking guidelines above, this was an interesting question with many viable approaches.
- The most common difficulty involved starting with

$$3^{2n} - 1 = (3 - 1)(3^{2n-1} + 3^{2n-2} + \dots + 3 + 1)$$

and then trying to show that

$$3^{2n-1} + 3^{2n-2} + \dots + 3 + 1 \text{ is even.}$$

Students who attempted to argue that powers of 3 are odd, and the sum of many odd numbers is even, were close and awarded 2 marks.

The crucial detail required for full marks is that the sum consists of $2n$ terms, and $2n$ is even. So the sum of an even number of odd terms is even, as required.

Question 3 (13 marks)

- a) On the Argand diagram, let A represents $z_1 = 1 + i\sqrt{3}$, let B represents $z_2 = 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$ and let C represents $z_3 = z_1 + z_2$. Find the exact coordinates of point C.

$$z_1 = 1 + i\sqrt{3}$$

$$z_2 = 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$$

$$= 2\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$$

$$= -1 + i\sqrt{3}$$

$$z_3 = z_1 + z_2$$

$$= 1 + i\sqrt{3} - 1 + i\sqrt{3}$$

$$\therefore z_3 = 2\sqrt{3}i$$

$$\text{hence } \underline{\underline{C = (0, 2\sqrt{3})}}$$

Students need to read questions carefully!

One mark was awarded for the correct coordinates.

b) Prove the following statement is false:

"There is a Pythagorean Triad where all numbers are odd."

2

It is sufficient to prove that a Pythagorean Triad does not exist where all numbers are odd.

Assume all numbers are odd, prove by contradiction:

Let the triad of odd numbers be a, b, c where
 $a = 2p+1, b = 2q+1, c = 2r+1, p, q, r \in \mathbb{Z}^+$

$$a^2 + b^2 = c^2$$

$$\text{LHS} = (2p+1)^2 + (2q+1)^2$$

$$= 4p^2 + 4p + 1 + 4q^2 + 4q + 1$$

$$= 4(p^2 + p + q^2 + q) + 2$$

$$= 2(2p^2 + 2p + 2q^2 + 2q + 1)$$

$$= 2m, m \in \mathbb{Z}^+ \quad \therefore \text{LHS is even}$$

$$\text{RHS} = (2r+1)^2$$

$$= 4r^2 + 4r + 1$$

$$= 4(r^2 + r) + 1$$

$$= 4n + 1, n \in \mathbb{Z}^+ \quad \therefore \text{RHS is odd}$$

Since LHS is even and RHS is odd

This is a contradiction

That is: $a^2 + b^2 = c^2$ does not exist where all numbers are odd.

Hence statement is false.

A single counterexample was not awarded any marks.

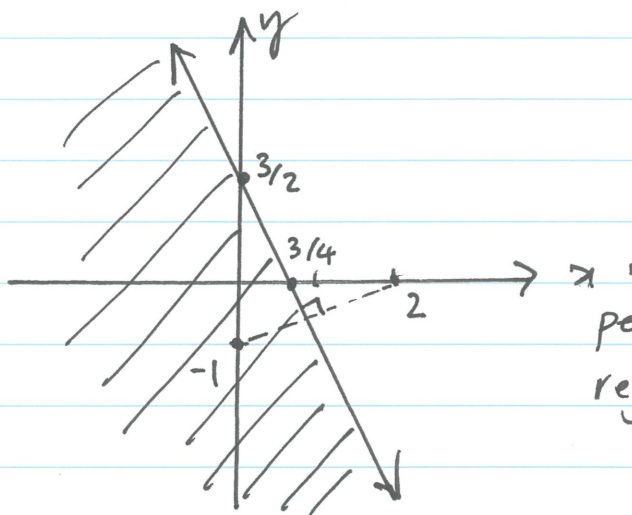
Students had to make a clear statement for one mark

c) Sketch the following regions: (show all working).

i) $|z+i| \leq |z-2| \Rightarrow |z-(0-i)| \leq |z-(2+0i)|$

2

$\Rightarrow$ perp. bisector of interval joining $(0, -1)$ and $(2, 0)$.

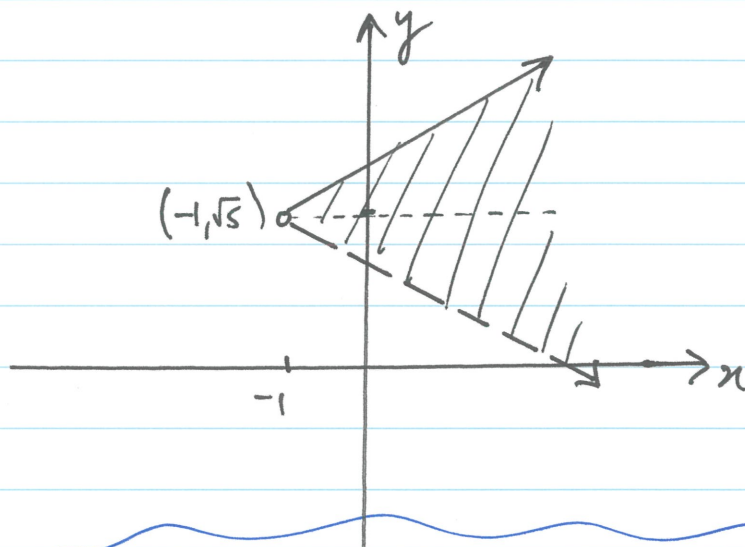


perp. solid line ✓
region below line ✓

ii) $-\frac{\pi}{4} < \text{Arg}(z+1-i\sqrt{5}) \leq \frac{\pi}{4}$

2

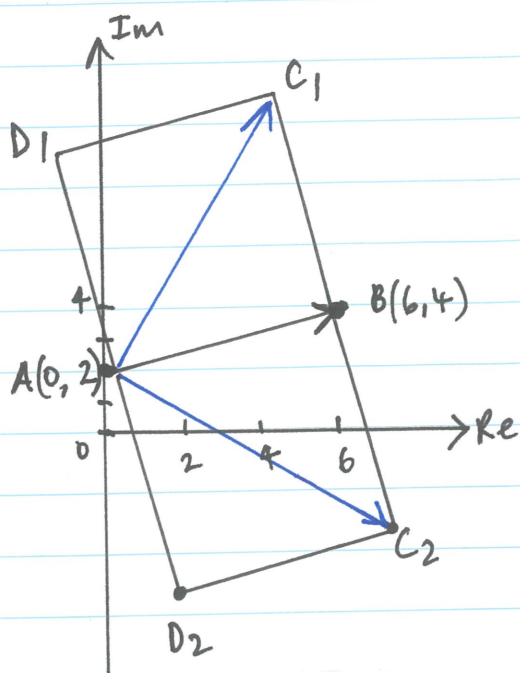
$-\frac{\pi}{4} < \text{Arg}(z - (-1 + \sqrt{5}i)) \leq \frac{\pi}{4}$



region between rays ✓
open circle at point $(-1, \sqrt{5})$.
solid / dotted rays. ✓

* (Care must be taken to determine the points of origin in part i) and ii)

- d) Let A and B be points representing $2i$ and $4i+6$ respectively.
Suppose AB is one side of a square $ABCD$. Find all the possible complex numbers representing C and D .



★ Marks were awarded for position vectors or points

★ A labelled diagram helped in marking student responses.

$$\begin{aligned}\vec{AB} &= 6 + 4i - 2i \\ &= 6 + 2i\end{aligned}$$

$\vec{AD}_1$ anticlockwise rotation by i

$$\begin{aligned}\therefore \vec{AD}_1 &= i \times \vec{AB} \\ &= i(6 + 2i) \\ &= -2 + 6i\end{aligned}$$

$$\begin{aligned}\text{point } D_1 &= \vec{OA} + \vec{AD}_1 \\ &= 0 + 2i - 2 + 6i \quad \checkmark \\ \therefore D_1 &= -2 + 8i = (-2, 8)\end{aligned}$$

$$\begin{aligned}\vec{AC}_1 &= \vec{AD}_1 + \vec{AB} \quad (\text{diagonal}) \\ &= -2 + 6i + 6 + 2i \\ &= 4 + 8i\end{aligned}$$

$$\begin{aligned}\text{point } C_1 &= \vec{OA} + \vec{AC}_1 \\ &= 0 + 2i + 4 + 8i \quad \checkmark \\ \therefore C_1 &= 4 + 10i = (4, 10)\end{aligned}$$

$\vec{AD}_2$ clockwise rotation by i

$$\begin{aligned}\therefore \vec{AD}_2 &= -i \times \vec{AB} \\ &= -i(6 + 2i) \\ &= 2 - 6i\end{aligned}$$

$$\begin{aligned}\text{point } D_2 &= \vec{OA} + \vec{AD}_2 \\ &= 0 + 2i + 2 - 6i\end{aligned}$$

$$\therefore D_2 = 2 - 4i = (2, -4)$$

$$\begin{aligned}\vec{AC}_2 &= \vec{AD}_2 + \vec{AB} \quad (\text{diagonal}) \\ &= 2 - 6i + 6 + 2i \\ &= 8 - 4i\end{aligned}$$

$$\begin{aligned}\text{point } C_2 &= \vec{OA} + \vec{AC}_2 \\ &= 0 + 2i + 8 - 4i\end{aligned}$$

$$\therefore C_2 = 8 - 2i = (8, -2)$$

e) Use proof by contradiction to show that $\frac{1}{\sqrt{2}}$ is irrational.

3

Assume $\frac{1}{\sqrt{2}}$ is rational. That is:

$\frac{1}{\sqrt{2}} = \frac{p}{q}$ where $p, q \in \mathbb{Z}^+$, $q \neq 0$, p, q have no common factors. ✓

$$\therefore \frac{1}{2} = \frac{p^2}{q^2}$$

$$q^2 = 2p^2 \dots \textcircled{1}$$

$\therefore q^2$ is even $\Rightarrow q$ is even

Let $q = 2k$ sub into $\textcircled{1}$ $k \in \mathbb{Z}^+$ ✓

$$(2k)^2 = 2p^2$$

$$4k^2 = 2p^2$$

$$p^2 = 2k^2$$

$\therefore p^2$ is even $\Rightarrow p$ is even

p, q have a common factor of 2.

Contradiction!

hence $\frac{1}{\sqrt{2}}$ is irrational. ✓

* A clear statement of proof and showing that p and q have a 'common factor' was required for 3 marks.

Question 4

(a) let abc be a three-digit number.

Suppose that $a+b+c = 9k$, $k \in \mathbb{Z}$

$$\begin{aligned} abc &= 100a + 10b + c \\ &= 100a + 10b + 9k - a - b \\ &= 99a + 9b + 9k \\ &= 9(11a + b + k) \\ &\quad \in \mathbb{Z} \text{ since } a, b, k \in \mathbb{Z} \end{aligned}$$

$\therefore abc$ is div. by 9. ✓✓ (1 M for correct progress*)

Now suppose that $100a + 10b + c = 9k$, $k \in \mathbb{Z}$

$$\begin{aligned} a+b+c &= a + b + 9k - 100a - 10b \\ &= 9k - 99a - 9b \\ &= 9(k - 11a - b) \\ &\quad \in \mathbb{Z} \text{ since } a, b, k \in \mathbb{Z} \end{aligned}$$

$\therefore a+b+c$ is div by 9. ✓

$\therefore$ Proposition is proven.

Comment

- Some students did not know that the three-digit number abc could be expressed as $100a + 10b + c$.
- Some students also did not prove the converse.

* correct progress : Writing $a+b+c = 9k$
or
 $100a + 10b + c = 9k$

$$(b) \quad \vec{OC} = \vec{OA} \times i \times \frac{1}{2}$$

$$= w \times \frac{1}{2} i$$

$$= \frac{1}{2} wi \quad \checkmark$$

$$\therefore D \text{ represents } \frac{w + \frac{1}{2} wi}{2}$$

$$= \frac{2w + wi}{4}$$

$$= \frac{w(2+i)}{4} \quad \checkmark$$

Comment

- Some students incorrectly wrote

$$\vec{OC} = 2wi \quad \text{or} \quad \vec{OC} = \frac{w}{2}$$

- Some students also thought that D can be found by finding $\frac{1}{2} \vec{AC}$.

$$\begin{aligned}
 (c) \quad & 4^n + 111(2^n) + 2024 \\
 &= (2^2)^n + 111(2^n) + 2024 \\
 &= (2^n)^2 + 111(2^n) + 2024 \\
 &= (2^n + 23)(2^n + 88) \quad \checkmark
 \end{aligned}$$

This is needed to show that the expression has factors other than 1 and itself, and hence it is composite.

Each factor is an integer > 1 since $n \in \mathbb{Z}^+$. $\checkmark$
 $\therefore 4^n + 111(2^n) + 2024$ is composite.

Method 2

$$\begin{aligned}
 & 4^n + 111(2^n) + 2024 \\
 &= 2^{2n} + 111(2^n) + 2024 \\
 &= 2 \left(2^{2n-1} + 111(2^{n-1}) + 1012 \right) \\
 &\quad \underbrace{\hspace{10em}}_{\in \mathbb{Z} \text{ with value } > 1} \quad \checkmark
 \end{aligned}$$

$\therefore 4^n + 111(2^n) + 2024$ is an even number that is greater than 2, so it is composite. $\checkmark$

required (students who stated that since it is even, it must be composite. Not true - counterexample: 2)

Comment

To achieve full marks, students had to have a careful justification of why the expression is composite.

See the above annotations.

(d)

(i) $z_1 = \text{cis } \alpha$

$$z_2 = k \text{cis } \beta$$

$$z_3 = (2-k) \text{cis } \gamma \quad \checkmark \checkmark$$

(ii) $z_1 + z_2 + z_3 = 0$

$$z_2 + z_3 = -z_1$$

$$\therefore |z_2 + z_3| = |z_1| \quad \checkmark$$

$$\therefore |k \cos \beta + (2-k) \cos \gamma + i(k \sin \beta + (2-k) \sin \gamma)| = 1$$

$$\left(k \cos \beta + (2-k) \cos \gamma \right)^2 + \left(k \sin \beta + (2-k) \sin \gamma \right)^2 = 1 \quad \checkmark$$

$$k^2 (\cos^2 \beta + \sin^2 \beta) + 2k(2-k) (\cos \beta \cos \gamma + \sin \beta \sin \gamma) + (2-k)^2 (\cos^2 \gamma + \sin^2 \gamma) = 1$$

$$\therefore k^2 + (2-k)^2 + 2k(2-k) \cos(\beta - \gamma) = 1 \quad \checkmark$$

Method 2

$$z_1 + z_2 + z_3 = 0$$

$$\text{cis } \alpha + k \text{cis } \beta + (2-k) \text{cis } \gamma = 0$$

Equating Re and Im parts:

$$\cos \alpha + k \cos \beta + (2-k) \cos \gamma = 0$$

$$\sin \alpha + k \sin \beta + (2-k) \sin \gamma = 0$$

} ✓

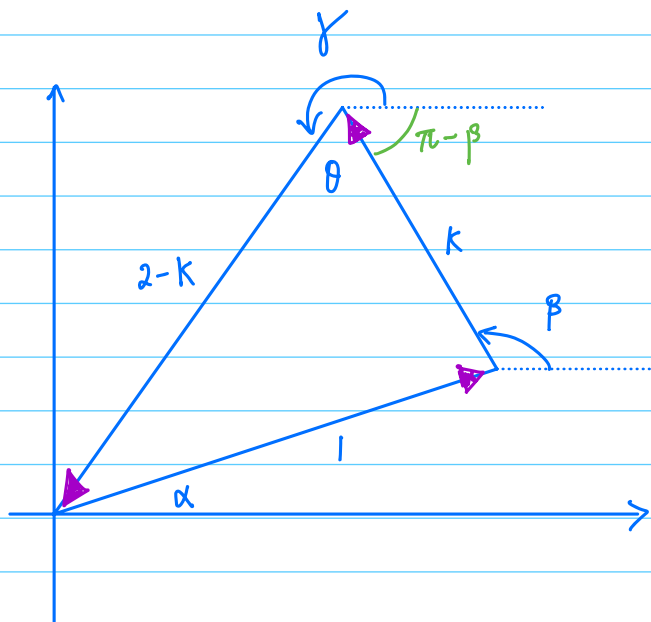
$$\therefore \begin{cases} k \cos \beta + (2-k) \cos \gamma = -\cos \alpha & (1) \\ k \sin \beta + (2-k) \sin \gamma = -\sin \alpha & (2) \end{cases}$$

$$(1)^2 + (2)^2:$$

$$k^2(\cos^2 \beta + \sin^2 \beta) + 2k(2-k)(\cos \beta \cos \gamma + \sin \beta \sin \gamma) + (2-k)^2(\cos^2 \gamma + \sin^2 \gamma) = \cos^2 \alpha + \sin^2 \alpha$$

$$\therefore k^2 + (2-k)^2 + 2k(2-k) \cos(\beta - \gamma) = 1 \quad \checkmark \checkmark$$

Method 3



$$\theta = 2\pi - (\pi - \beta) - \gamma$$

$$= \pi + \beta - \gamma$$

$$\therefore 1^2 = k^2 + (2-k)^2 - 2k(2-k) \cos(\pi + \beta - \gamma)$$

$$1 = k^2 + (2-k)^2 - 2k(2-k)(-\cos(\beta - \gamma)) \quad (\cos(\pi + A) = -\cos A)$$

$$\therefore k^2 + (2-k)^2 + 2k(2-k) \cos(\beta - \gamma) = 1$$

Comment

Not well done.

Students who resorted to the cosine rule needed a careful diagram. Incorrect diagrams or using the cosine rule incorrectly were penalised.